Maastricht University

Department of Advanced Computing Sciences

Lecture 4: The Transportation Problem

[based on slides by: Steven Kelk]

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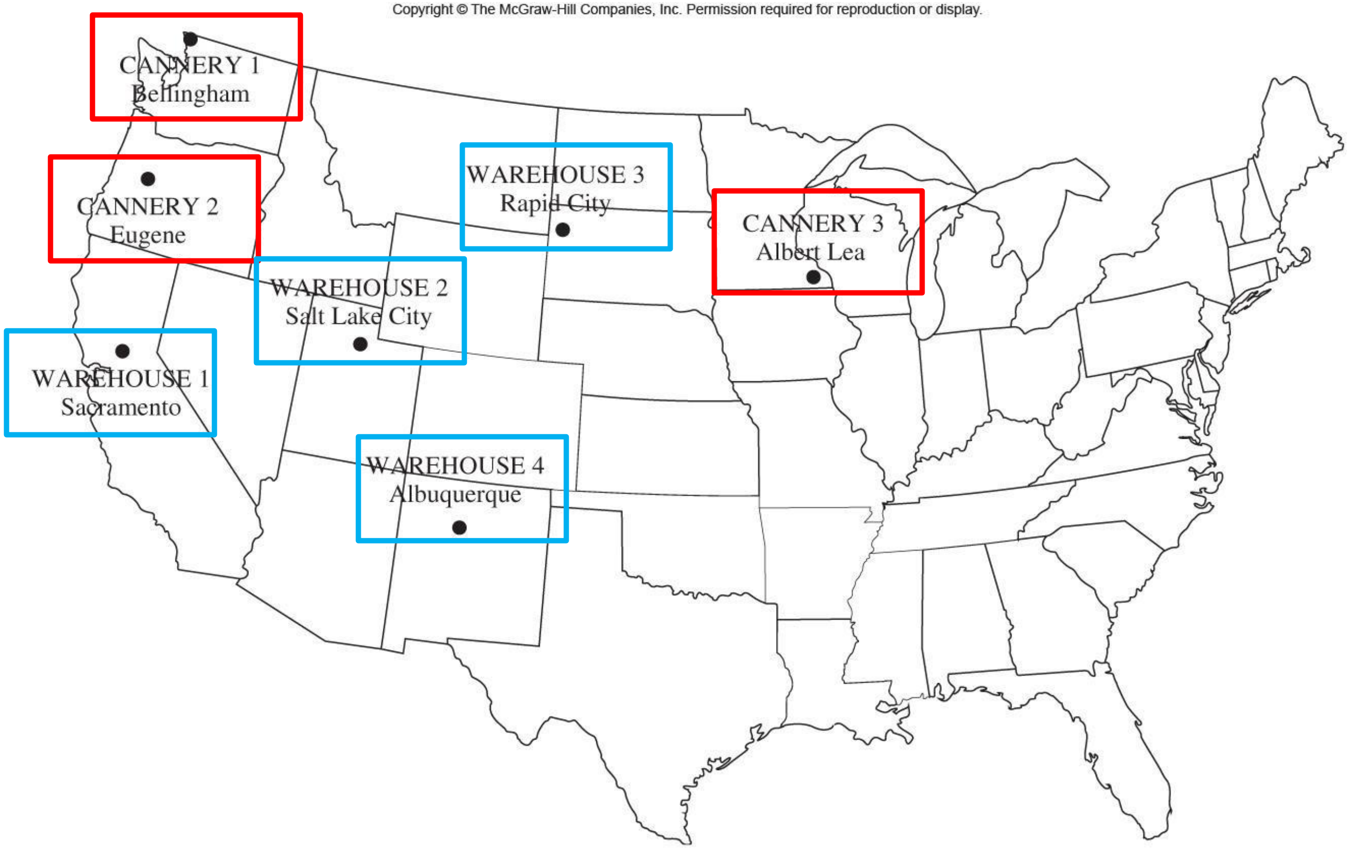
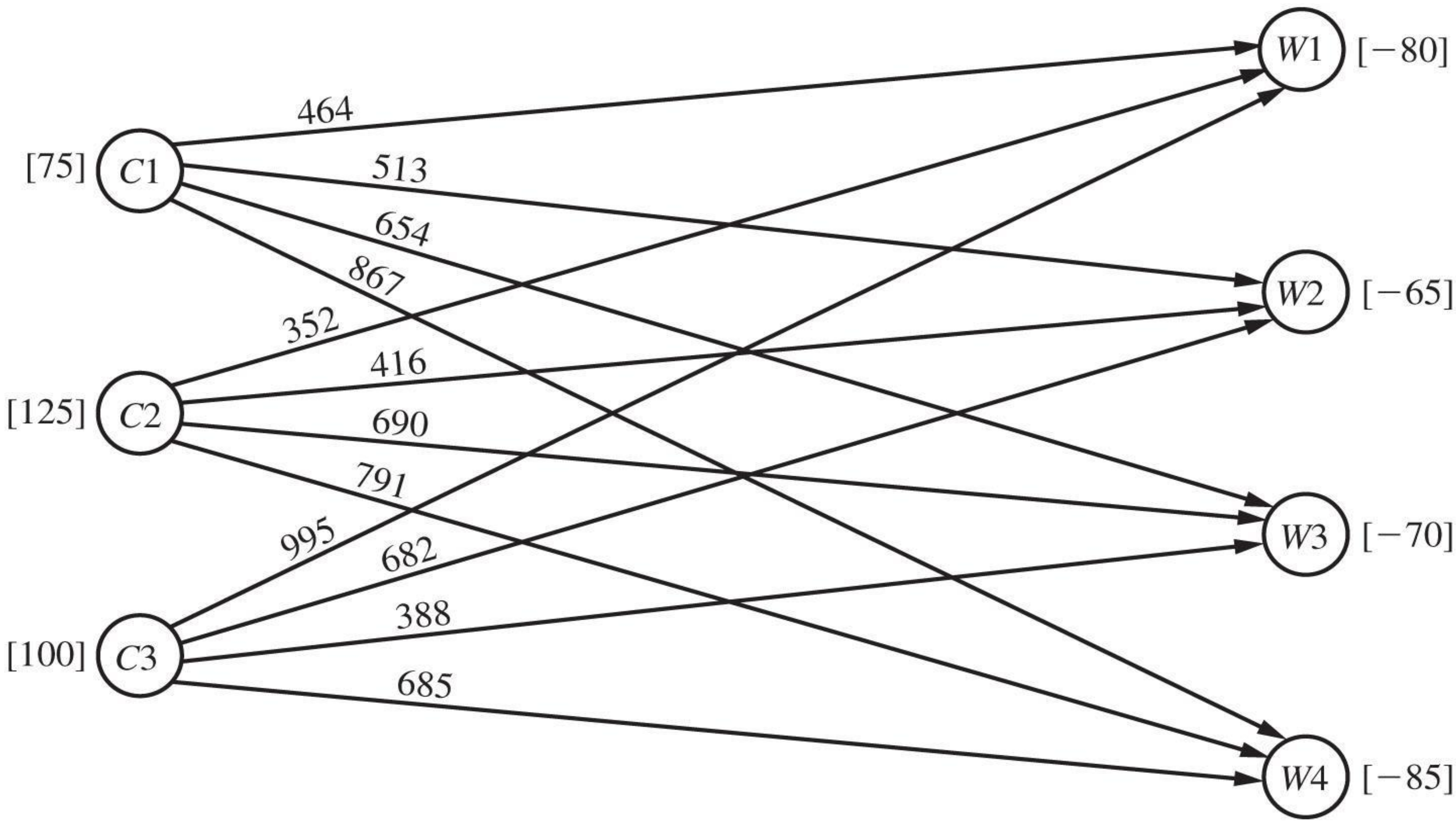
P&T's canned peas



- There are 300 truckloads to be shipped
- We have to plan how to assign the shipments to the canneries-warehouses combinations while minimizing the shipment costs

P&T's canned peas



P&T's canned peas

TABLE 8.2 Shipping data for P & T Co.

		Shipping Cost ($) per Truckload				
		Warehouse				
		1	2	3	4	Output
Cannery	1	464	513	654	867	75
	2	352	416	690	791	125
	3	995	682	388	685	100
Allocation		80	65	70	85	

Transportation costs

Needs of warehouses

Production of canneries =300

How would you write the LP problem?

P&T's canned peas

$$\text{Minimize} \quad Z = 464x_{11} + 513x_{12} + 654x_{13} + 867x_{14} + 352x_{21} + 416x_{22} + 690x_{23} + 791x_{24} + 995x_{31} + 682x_{32} + 388x_{33} + 685x_{34},$$

subject to the constraints

$$\begin{array}{llllllllllllllll}
x_{11} + x_{12} + x_{13} + x_{14} & & & & = 75 \\
& x_{21} + x_{22} + x_{23} + x_{24} & & & = 125 \\
& & x_{31} + x_{32} + x_{33} + x_{34} & & = 100 \\
x_{11} & + x_{21} & + x_{31} & & = 80 \\
x_{12} & + x_{22} & + x_{32} & & = 65 \\
x_{13} & + x_{23} & + x_{33} & & = 70 \\
x_{14} & + x_{24} & + x_{34} & & = 85
\end{array}$$

and

$$x_{ij} \geq 0 \qquad (i = 1, 2, 3; j = 1, 2, 3, 4).$$

Do you see something interesting here?

P&T's canned peas

TABLE 8.3 Constraint coefficients for P & T Co.

Coefficient of:

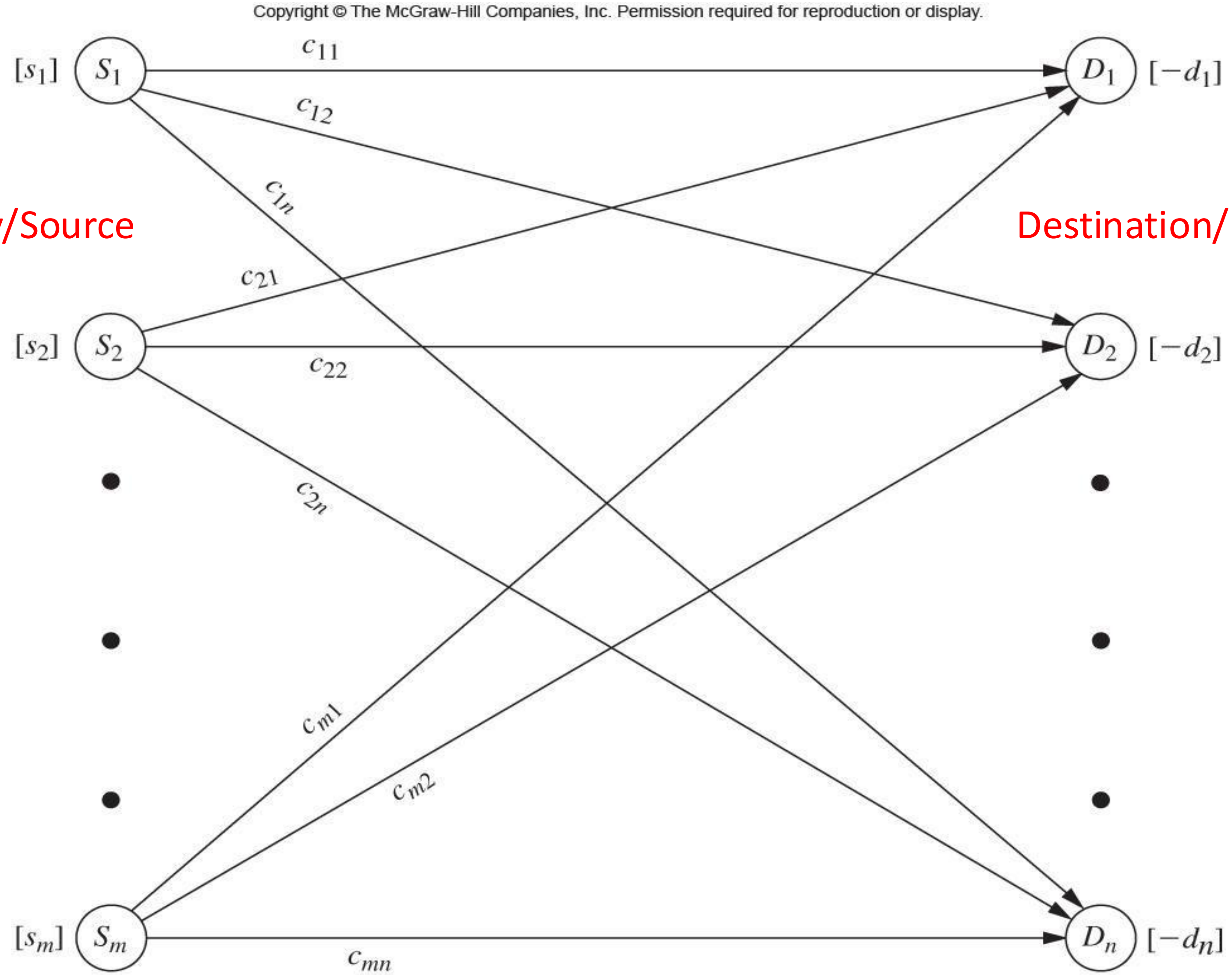
$$
\begin{array}{cc}
 & \begin{matrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{21} & x_{22} & x_{23} & x_{24} & x_{31} & x_{32} & x_{33} & x_{34} \end{matrix} \\
\mathbf{A} = & \left[\begin{matrix}
1 & 1 & 1 & 1 & & & & & & & & \\
 & & & & 1 & 1 & 1 & 1 & & & & \\
 & & & & & & & & 1 & 1 & 1 & 1 \\
1 & & & & 1 & & & & 1 & & & \\
 & 1 & & & & 1 & & & & 1 & & \\
 & & 1 & & & & 1 & & & & 1 & \\
 & & & 1 & & & & 1 & & & & 1
\end{matrix}\right]
\begin{matrix}
\left.\vphantom{\begin{matrix}1\\1\\1\end{matrix}}\right\} \text{Cannery constraints} \\
\left.\vphantom{\begin{matrix}1\\1\\1\\1\end{matrix}}\right\} \text{Warehouse constraints}
\end{matrix}
\end{array}
$$

The model can be generalized to every source-destination problem

General case

Supply/Source

Destination/Demand



General case

TABLE 8.5 Parameter table for the transportation problem

	Cost per Unit Distributed				
	Destination				
	1	**2**	⋯	***n***	**Supply**
Source 1	c_{11}	c_{12}	⋯	c_{1n}	s_1
2	c_{21}	c_{22}	⋯	c_{2n}	s_2
⋮	⋯	⋯	⋯	⋯	⋮
m	c_{m1}	c_{m2}	⋯	c_{mn}	s_m
Demand	d_1	d_2	⋯	d_n	

Sum of demands must equal sum of supplies!

General case

TABLE 8.6 Constraint coefficients for the transportation problem

Coefficient of:

$$
\begin{array}{c}
\begin{array}{cccccccccccccc}
x_{11} & x_{12} & \cdots & x_{1n} & x_{21} & x_{22} & \cdots & x_{2n} & \cdots & x_{m1} & x_{m2} & \cdots & x_{mn}
\end{array}
\\[6pt]
\mathbf{A} = \left[\begin{array}{ccccccccccccc}
1 & 1 & \cdots & 1 & & & & & & & & & \\
 & & & & 1 & 1 & \cdots & 1 & & & & & \\
 & & & & & & & & \ddots & & & & \\
 & & & & & & & & & 1 & 1 & \cdots & 1 \\
1 & & & & 1 & & & & & 1 & & & \\
 & 1 & & & & 1 & & & \cdots & & 1 & & \\
 & & \ddots & & & & \ddots & & & & & \ddots & \\
 & & & 1 & & & & 1 & & & & & 1
\end{array}\right]
\begin{array}{l}
\left.\begin{array}{c} \\ \\ \\ \\ \end{array}\right\} \text{Supply constraints} \\
\left.\begin{array}{c} \\ \\ \\ \\ \end{array}\right\} \text{Demand constraints}
\end{array}
\end{array}
$$

(m+n) constraints corresponding to m sources (i.e. supply points) and n destinations (i.e. demand points)

General case

$$\text{Minimize} \qquad Z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij},$$

subject to

Source constraints

$$\sum_{j=1}^{n} x_{ij} = s_i \qquad \text{for } i = 1, 2, \ldots, m,$$

Demand constraints

$$\sum_{i=1}^{m} x_{ij} = d_j \qquad \text{for } j = 1, 2, \ldots, n,$$

and

$$x_{ij} \geq 0, \qquad \text{for all } i \text{ and } j.$$

Relaxation

The sum of supplies = sum of demands restriction can be softened

⇨ Use *dummy destinations* and/or *dummy sources*.

This increases the modelling power of the Transportation Problem considerably.

Dummy Destination

TABLE 8.7 Production scheduling data for Northern Airplane Co.

Month	Scheduled Installations	Maximum Production	Unit Cost* of Production	Unit Cost* of Storage
1	10	25	1.08	0.015
2	15	35	1.11	0.015
3	25	30	1.10	0.015
4	20	10	1.13	

Cumulative number of engines produced in a month

There might be an advantage in producing and storing, i.e., to save money you are allowed to produce engines in month x that are only going to be fitted in month $y > x$, but each such engine will have to be stored for $(y-x)$ months.

How to model this as a Transportation Problem?
(And how to prevent fractional engine production…?)

Dummy Destination

TABLE 8.8 Incomplete parameter table for Northern Airplane Co.

		Cost per Unit Distributed				
		Destination				
		1	2	3	4	Supply
Source	1					
	2					
	3					
	4					
Demand						

Dummy Destination

TABLE 8.8 Incomplete parameter table for Northern Airplane Co.

		Cost per Unit Distributed				
		Destination (month)				
		1	2	3	4	Supply
Source (month)	1	1.080	1.095	1.110	1.125	?
	2	?	1.110	1.125	1.140	?
	3	?	?	1.100	1.115	?
	4	?	?	?	1.130	?
Demand		10	15	25	20	

- Let $x_{i,j}$ be the number of engines produced in month *i* for installation in month *j*
- Fill the costs: $c_{1,3} = 1.110 = 1.080 + ((3\text{-}1)*0.015)$.
- But what to do about the positions marked with a "?" symbol?

Filling the blanks

TABLE 8.9 Complete parameter table for Northern Airplane Co.

		Cost per Unit Distributed					
		Destination					
		1	**2**	**3**	**4**	**5(*D*)**	**Supply**
Source	1	1.080	1.095	1.110	1.125		
	2	*M*	1.110	1.125	1.140		
	3	*M*	*M*	1.100	1.115		
	4	*M*	*M*	*M*	1.130		
Demand		10	15	25	20		

- We cannot produce after the scheduled intervention
- **Big M** technique to make it too expensive to produce for the past
- How do we fix the supply?

Filling the blanks

TABLE 8.9 Complete parameter table for Northern Airplane Co.

		Cost per Unit Distributed					
		Destination					
		1	**2**	**3**	**4**	**5(*D*)**	**Supply**
Source	1	1.080	1.095	1.110	1.125	0	25
	2	*M*	1.110	1.125	1.140	0	35
	3	*M*	*M*	1.100	1.115	0	30
	4	*M*	*M*	*M*	1.130	0	10
Demand		10	15	25	20		

- Set month supply to maximum month production
- Use a **dummy destination** to absorb “fictional” production

Filling the blanks

TABLE 8.9 Complete parameter table for Northern Airplane Co.

		Cost per Unit Distributed					
		Destination					
		1	**2**	**3**	**4**	**5(*D*)**	**Supply**
Source	1	1.080	1.095	1.110	1.125	0	25
	2	*M*	1.110	1.125	1.140	0	35
	3	*M*	*M*	1.100	1.115	0	30
	4	*M*	*M*	*M*	1.130	0	10
Demand		10	15	25	20	30	

- How did we determine the demand of the dummy destination?
- Total unused capacity = sum of supply – sum of demand (25+35+30+10)-(10+15+25+20)=30

Big M technique

TABLE 8.9 Complete parameter table for Northern Airplane Co.

		Cost per Unit Distributed					
		Destination					
		1	**2**	**3**	**4**	**5(*D*)**	**Supply**
Source	1	1.080	1.095	1.110	1.125	0	25
	2	*M*	1.110	1.125	1.140	0	35
	3	*M*	*M*	1.100	1.115	0	30
	4	*M*	*M*	*M*	1.130	0	10
Demand		10	15	25	20	30	

- How do we prevent fractional production of energy?
- **Magical property of the Transportation Problem**: if the sources and the demands are all integers, there will exist an integer optimal solution.
- **Every** Basic Feasible Solution will be integer!

Dummy sources

TABLE 8.10 Water resources data for Metro Water District

	Cost (Tens of Dollars) per Acre Foot				
	Berdoo	**Los Devils**	**San Go**	**Hollyglass**	**Supply**
Colombo River	16	13	22	17	50
Sacron River	14	13	19	15	60
Calorie River	19	20	23	—	50
Minimum needed	30	70	0	10	(in units of 1
Requested	50	70	30	∞	million acre feet)

- We want to supply the 4 cities with water from the 3 rivers.
- Hollyglass cannot be supplied from Calorie River.
- Cities have minimum water requirements (*"minimum needed"*).
- Cities have maximum water requirements (*"requested"*).

Dummy sources

TABLE 8.10 Water resources data for Metro Water District

	Cost (Tens of Dollars) per Acre Foot				
	Berdoo	**Los Devils**	**San Go**	**Hollyglass**	**Supply**
Colombo River	16	13	22	17	50
Sacron River	14	13	19	15	60
Calorie River	19	20	23	—	50
Minimum needed	30	70	0	10	(in units of 1
Requested	50	70	30	∞	million acre feet)

- Note that all the water in the rivers has to be shipped
- Note that Hollyglass will take as much as it can get!

Dummy sources

TABLE 8.10 Water resources data for Metro Water District

	Cost (Tens of Dollars) per Acre Foot				
	Berdoo	**Los Devils**	**San Go**	**Hollyglass**	**Supply**
Colombo River	16	13	22	17	50
Sacron River	14	13	19	15	60
Calorie River	19	20	23	—	50
Minimum needed	30	70	0	10	(in units of 1
Requested	50	70	30	60	million acre feet)

- Note that all the water in the rivers has to be shipped
- Note that Hollyglass will take as much as it can get!
  Requested demand for Hollyglass: exceeded total supply – minimum need of other cities
  (50+60+50)-(30+70+0)=60

Dummy sources

TABLE 8.11 Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	Supply
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	$\boxed{M}$	50
Demand		50	70	30	60	

Calorie River cannot supply Hollyglass

Dummy sources

TABLE 8.11 Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	Supply
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	*M*	50
Demand		50	70	30	60	

- We have an ECCESS DEMAND CAPACITY!
- Supply = sum of the requested amount – total real supply (50+70+30+60)-(50+60+50)=50

Dummy sources

TABLE 8.11 Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	Supply
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	*M*	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

- We have an ECCESS DEMAND CAPACITY!
- Supply = sum of the requested amount – total real supply (50+70+30+60)-(50+60+50)=50
- Hence, dummy source!
  No costs (it is not a real source)

Dummy sources

TABLE 8.11 Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	Supply
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	*M*	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

- The table shown above is a *partial* solution to the problem. It correctly models the upper bound on consumption by a city (the "requirement"), allows cities to receive less than their requirement, and prevents Hollyglass from consuming Calorie River water.
- What about the lower bound…?

Dummy sources

TABLE 8.11 Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	Supply
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	*M*	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	
Min Needed:		30	70	0	10	

- This is a fascinating example because for each city we can use a different technique for modelling its lower bound.

TABLE 8.11 Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	Supply
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	*M*	50
	Dummy	0	0	0	0	50
Demand		50	70	30	60	

Min Needed:	30	70	0	10

- San Go: Easy! San Go has no minimum requirement.

TABLE 8.11 Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	**Los Devils**	**San Go**	**Hollyglass**	**Supply**
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	*M*	50
	Dummy	0	$\boxed{0}$	0	0	50
Demand		50	70	30	60	
Min Needed:		30	70	0	10	

- San Go: Easy! San Go has no minimum requirement.
- Los Devils: Use Big M to forbid *any* dummy consumption

TABLE 8.11 Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	Supply
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	*M*	50
	Dummy	0	$\boxed{0}$	0	0	50
Demand		50	70	30	60	
Min Needed:		30	70	0	10	

- San Go: Easy! San Go has no minimum requirement.
- Los Devils: Use Big M to forbid *any* dummy consumption
- Hollyglass: We don't need to change anything.
  (receives 50 from dummy source hence it will get at least 10 from real ones)

TABLE 8.11 Parameter table without minimum needs for Metro Water District

		Cost (Tens of Millions of Dollars) per Unit Distributed				
		Destination				
		Berdoo	Los Devils	San Go	Hollyglass	Supply
Source	Colombo River	16	13	22	17	50
	Sacron River	14	13	19	15	60
	Calorie River	19	20	23	*M*	50
	Dummy	0	$\boxed{0}$	0	0	50
Demand		50	70	30	60	
Min Needed:		30	70	0	10	

- San Go: Easy! San Go has no minimum requirement.
- Los Devils: Use Big M to forbid *any* dummy consumption
- Hollyglass: We don't need to change anything.
- Berdoo: we need a slightly more sophisticated technique…

TABLE 8.12 Parameter table for Metro Water District

			Cost (Tens of Millions of Dollars) per Unit Distributed					
			Destination					
			Berdoo (min.) 1	Berdoo (extra) 2	Los Devils 3	San Go 4	Hollyglass 5	Supply
Source	Colombo River	1	16	16	13	22	17	50
	Sacron River	2	14	14	13	19	15	60
	Calorie River	3	19	19	20	23	*M*	50
	Dummy	4(*D*)	*M*	0	*M*	0	0	50
Demand			30	20	70	30	60	

- We split Berdoo into **two** destinations.
- The first part demands the minimum requirement (30 units) and refuses to accept "dummy" water, like Los Devils.
- The second part accepts any surplus above the minimum, up to a maximum of 20 (because 30+20=50), like San Go.

Why the TSM?

- The Transportation Problem can easily be modelled by a normal LP, and we know how to solve LPs already (Simplex Method), so why do we bother with a special formulation? And why do we bother trying to squeeze complex models into a format suitable for the Transportation Problem?

- **Speed.** We can solve it **much faster** than using a standard LP solver. Especially important for very big problems!

- **Insight.** We know a huge amount about Transportation Problems, (magical **integer solutions property**,) so no need for Integer Linear Programming. (It also has various nice theoretical properties which are not known to hold for general Linear Programs).

Transportation Simplex Method

Conceptually, the Transportation Simplex Method is very similar to the general Simplex Method:

1. Construct an **initial** Basic Feasible Solution (BFS).

2. Check whether the current BFS is **optimal**.
   YES: we are done!

3. NO: **Identify** a *nonbasic* variable with the most potential to improve the objective function. This is the *entering* variable. **Identify** a variable in the current basis that will *leave* when the entering variable enters

4. This gives us a new BFS; go back to 2.

Transportation Simplex Method

Fortunately, the Transportation Simplex Method (TSM) has some shortcuts:

1. Construct an **initial** Basic Feasible Solution (BFS).

    - For the general Simplex Method this can be hard work, because we have to introduce artificial variables and pivot multiple times to drive them to zero (i.e. to enter the feasible region).

    - With the TSM we can find an initial BFS very easily (there are rules)

Transportation Simplex Method

Fortunately, the Transportation Simplex Method (TSM) has some shortcuts:

1. Construct an **initial** Basic Feasible Solution (BFS).
2. Check whether the current BFS is **optimal**.
   YES: we are done!

   - In the general Simplex Method, we can determine this by looking at the top row of the Simplex Tableaux (no negative numbers in the top row).
   - Exactly the same idea works with TSM, the difference is simplicity: we can quickly and dynamically generate the top row by deriving a very simple set of linear equations from the current BFS and solving them.

Transportation Simplex Method

- Fortunately, the Transportation Simplex Method (TSM) has some shortcuts:

1. Construct an **initial** Basic Feasible Solution (BFS).
2. Check whether the current BFS is **optimal**.
   YES: we are done!
3. NO: **Identify** entering basic variable and leaving basic variable

   Just like the normal Simplex method, we choose the nonbasic variable with the most negative "top-row entry" as the entering basic variable.

Transportation Simplex Method

Fortunately, the Transportation Simplex Method (TSM) has some shortcuts:

1. Construct an **initial** Basic Feasible Solution (BFS).
2. Check whether the current BFS is **optimal**.
   YES: we are done!
3. NO: **Identify** entering basic variable and leaving basic variable

   - With the general Simplex Method, we identify the leaving basic variable with the Minimum Ratio Test, and then pivot.

   - With TSM we instead look for "closed loops" in a specially modified Transportation tableaux to find the leaving basic variable. The subsequent pivot can then be performed more quickly than in a general Simplex tableaux.

Table for the TSM

TABLE 8.13 Original simplex tableau before simplex method is applied to transportation problem

Basic Variable	Eq.	Coefficient of:								Right side	
		Z	$\cdots$	x_{ij}	$\cdots$	z_i	$\cdots$	z_{m+j}	$\cdots$		
Z	(0)	-1		c_{ij}		$\boxed{M}$		$\boxed{M}$		0	
	(1)										
	$\vdots$										
z_i	(i)	0		1		1				s_i	Supply
	$\vdots$										
z_{m+j}	$(m+j)$	0		1				1		d_j	Demand
	$\vdots$										
	$(m+n)$										

Table for the TSM

TABLE 8.14 Row 0 of simplex tableau when simplex method is applied to transportation problem

Basic Variable	Eq.	Coefficient of:								Right Side
		Z	$\cdots$	x_{ij}	$\cdots$	z_i	$\cdots$	z_{m+j}	$\cdots$	
Z	(0)	-1		$c_{ij} - u_i - v_j$		$M + \boxed{u_i}$		$M + \boxed{v_j}$		$-\sum_{i=1}^{m} s_i u_i - \sum_{j=1}^{n} d_j v_j$

u_i = multiple of original row(i) that has been subtracted from the original row(0) during the iterations up to this one

v_j = multiple of original row(m+j) that has been subtracted from the original row(0) during the iterations up to this one

TABLE 8.15 Format of a transportation simplex tableau

		Destination				Supply	u_i
		1	**2**	...	***n***		
Source	1	c_{11}	c_{12}	...	c_{1n}	s_1	
	2	c_{21}	c_{22}	...	c_{2n}	s_2	
	⋮	...	...	...	...	⋮	
	m	c_{m1}	c_{m2}	...	c_{mn}	s_m	
Demand		d_1	d_2	...	d_n	$Z =$	
	v_j						

Table 8.15

TABLE 8.15 Format of a transportation simplex tableau

		Destination					
		1	**2**	…	***n***	**Supply**	u_i
Source	1	c_{11}	c_{12}	…	c_{1n}	s_1	
	2	c_{21}	c_{22}	…	c_{2n}	s_2	
	⋮	…	…	…	…	⋮	
	m	c_{m1}	c_{m2}	…	c_{mn}	s_m	
Demand		d_1	d_2	…	d_n	$Z =$	
	v_j						

Additional information to be added to each cell:

If x_{ij} is a basic variable	If x_{ij} is a nonbasic variable
c_{ij} x_{ij}	c_{ij} $c_{ij} - u_i - v_j$

value of basic variable

"top-row" entry of corresponding nonbasic variable.

Table 8.15

TABLE 8.15 Format of a transportation simplex tableau

		Destination					
		1	**2**	⋯	***n***	**Supply**	u_i
Source	1	c_{11}	c_{12}	⋯	c_{1n}	s_1	
	2	c_{21}	c_{22}	⋯	c_{2n}	s_2	
	⋮	⋯	⋯	⋯	⋯	⋮	
	m	c_{m1}	c_{m2}	⋯	c_{mn}	s_m	
Demand		d_1	d_2	⋯	d_n	$Z =$	
	v_j						

Additional information to be added to each cell:

If x_{ij} is a basic variable

c_{ij}	
	x_{ij}

If x_{ij} is a nonbasic variable

c_{ij}	
	$c_{ij} - u_i - v_j$

*There will always be **(m+n-1)** basic variables.*

Transportation Simplex Method

1. Construct the trasportation table

Metro Water District

TABLE 8.12 Parameter table for Metro Water District

			Cost (Tens of Millions of Dollars) per Unit Distributed					
			Destination					
			Berdoo (min.) 1	**Berdoo (extra) 2**	**Los Devils 3**	**San Go 4**	**Hollyglass 5**	**Supply**
Source	Colombo River	1	16	16	13	22	17	50
	Sacron River	2	14	14	13	19	15	60
	Calorie River	3	19	19	20	23	*M*	50
	Dummy	4(*D*)	*M*	0	*M*	0	0	50
Demand			30	20	70	30	60	

Metro Water District

TABLE 8.16 Initial BF solution from the Northwest Corner Rule

		Destination 1	2	3	4	5	Supply	u_i
Source	1	16	16	13	22	17	50	
	2	14	14	13	19	15	60	
	3	19	19	20	23	*M*	50	
	4(*D*)	*M*	0	*M*	0	0	50	
Demand		30	20	70	30	60		
	v_j							

Transportation Simplex Method

1. Construct the table

2. Construct an initial Basic Feasible Solution (BFS).
   - North west corner rule
   - Make basic variable large enough to meet supply or demand.

Transportation Simplex Method

TABLE 8.16 Initial BF solution from the Northwest Corner Rule

		Destination						
		1	**2**	**3**	**4**	**5**	**Supply**	u_i
Source	1	16	16	13	22	17	50	
	2	14	14	13	19	15	60	
	3	19	19	20	23	M	50	
	4(D)	M	0	M	0	0	50	
Demand		30	20	70	30	60		
	v_j							

Transportation Simplex Method

1. Construct the table

2. Construct an initial Basic Feasible Solution (BFS).
   - North west corner rule
   - Make basic variable large enough to meet supply or demand.
   - Select remaining variables to be basic with the remaining feasible allocations

Transportation Simplex Method

TABLE 8.16 Initial BF solution from the Northwest Corner Rule

		Destination					Supply	u_i
		1	2	3	4	5		
Source	1	16	16	13	22	17	50	
	2	14	14	13	19	15	60	
	3	19	19	20	23	*M*	50	
	4(*D*)	*M*	0	*M*	0	0	50	
Demand		30	20	70	30	60		
	v_j							

Transportation Simplex Method

TABLE 8.16 Initial BF solution from the Northwest Corner Rule

		Destination 1	Destination 2	Destination 3	Destination 4	Destination 5	Supply	u_i
Source	1	16 (30)	16 (20)	13	22	17	50	
	2	14	14 (0)	13 (60)	19	15	60	
	3	19	19	20 (10)	23 (30)	M (10)	50	
	4(D)	M	0	M	0	0 (50)	50	
Demand		30	20	70	30	60	$Z = 2{,}470 + 10M$	
	v_j							

16*30+16*20+14*0+13*60+20*10+23*30+M*10

Transportation Simplex Method

1. Construct the table

2. Construct an initial Basic Feasible Solution (BFS).
   - North west corner rule
   - Make basic variable large enough to meet supply or demand.
   - Select remaining variables to be basic with the remaining feasible allocations

3. Determine whether we have reached optimality

Transportation Simplex Method

TABLE 8.16 **Initial BF solution from the Northwest Corner Rule**

		Destination						
		1	**2**	**3**	**4**	**5**	**Supply**	u_i
Source	1	16 30	16 20	13 ?	22 ?	17 ?	50	
	2	14 ?	14 0	13 60	19 ?	15 ?	60	
	3	19 ?	19 ?	20 10	23 30	M 10	50	
	4(*D*)	M ?	0 ?	M ?	0 ?	0 50	50	
Demand		30	20	70	30	60	$Z = 2{,}470 + 10M$	
	v_j							

Fill in the red question marks for all the nonbasic variables.

If none of the values we fill in are negative, we have reached optimality.

Transportation Simplex Method

TABLE 8.16 **Initial BF solution from the Northwest Corner Rule**

		Destination 1	2	3	4	5	Supply	u_i
Source	1	16 (30)	16 (20)	13 ?	22 ?	17 ?	50	u_1
	2	14 ?	14 (0)	13 (60)	19 ?	15 ?	60	
	3	19 ?	19 ?	20 (10)	23 (30)	*M* (10)	50	
	4(*D*)	*M* ?	0 ?	*M* ?	0 ?	0 (50)	50	
Demand		30	20	70	30	60	$Z = 2{,}470 + 10M$	
	v_j				v_4			

For each cell they are given by the equation $c_{ij} - u_i - v_j$ For example, we would replace the "?" in cell (1,4) with $22 - u_1 - v_4$

Transportation Simplex Method

TABLE 8.16 **Initial BF solution from the Northwest Corner Rule**

		Destination						
		1	**2**	**3**	**4**	**5**	**Supply**	u_i
Source	1	16 30	16 20	13 ?	22 ?	17 ?	50	
	2	14 ?	14 0	13 60	19 ?	15 ?	60	
	3	19 ?	19 ?	20 10	23 30	*M* 10	50	
	4(*D*)	*M* ?	0 ?	*M* ?	0 ?	0 50	50	
Demand		30	20	70	30	60	$Z = 2{,}470 + 10M$	
	v_j							

So we need to know the values of u_i and v_j !

Fortunately these can be derived from the current BFS (i.e., the circled guys).

Filling the blanks

- The idea is as follows. We need to compute the u_i and v_j values to be able to fill in the missing ? values.

- For the (m+n-1) basic variables, $c_{ij} – u_i – v_j = 0$ holds.

- So, by looking at the basic variables, we obtain (m+n –1) equations in (m+n) unknowns.

- We have one degree of freedom; we can kill this degree of freedom by arbitrarily setting one of the u_i and v_j values to zero. Then we can get a solution for the remaining u_i and v_j values by solving the remaining two-variable equations.

Transportation Simplex Method

TABLE 8.16 Initial BF solution from the Northwest Corner Rule

		Destination						
		1	**2**	**3**	**4**	**5**	**Supply**	u_i
Source	1	16 30	16 20	13	22	17	50	
	2	14	14 0	13 60	19	15	60	
	3	19	19	20 10	23 30	*M* 10	50	
	4(*D*)	*M*	0	*M*	0	0 50	50	
Demand		30	20	70	30	60	$Z = 2{,}470 + 10M$	
	v_j							

Transportation Simplex Method

1. Construct the table

2. Construct an initial Basic Feasible Solution (BFS).
   - North west corner rule
   - Make basic variable large enough to meet supply or demand.
   - Select remaining variables to be basic with the remaining feasible allocations

3. Determine whether we have reached optimality

4. Identify entering basic variable and leaving basic variable, then pivot.

Find EBV and LBV

- We choose the **most negative** value (for $c_{ij} – u_i – v_j$).
  This is just like choosing the column with the most negative top-row value in the normal Simplex Tableaux.

- This will become the **entering** variable.
  At the moment it is nonbasic, (i.e. equal to zero).
  We will make it basic (i.e. potentially larger than zero) and this will force a currently basic variable to **leave** and become nonbasic.

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 / 30	16 / 20	13 / -2	22 / 4	17 / -978	50	-5
S2	14 / 0	14 / 0	13 / 60	19 / 3	15 / -978	60	-7
S3	19 / -2	19 / -2	20 / 10	23 / 30	1000 / 10	50	0
S4	1000 / 1979	0 / 979	1000 / 1980	0 / 977	0 / 50	50	-1000
Demand	30	20	70	30	60	Z =	12470
v(j)	21	21	20	23	1000		

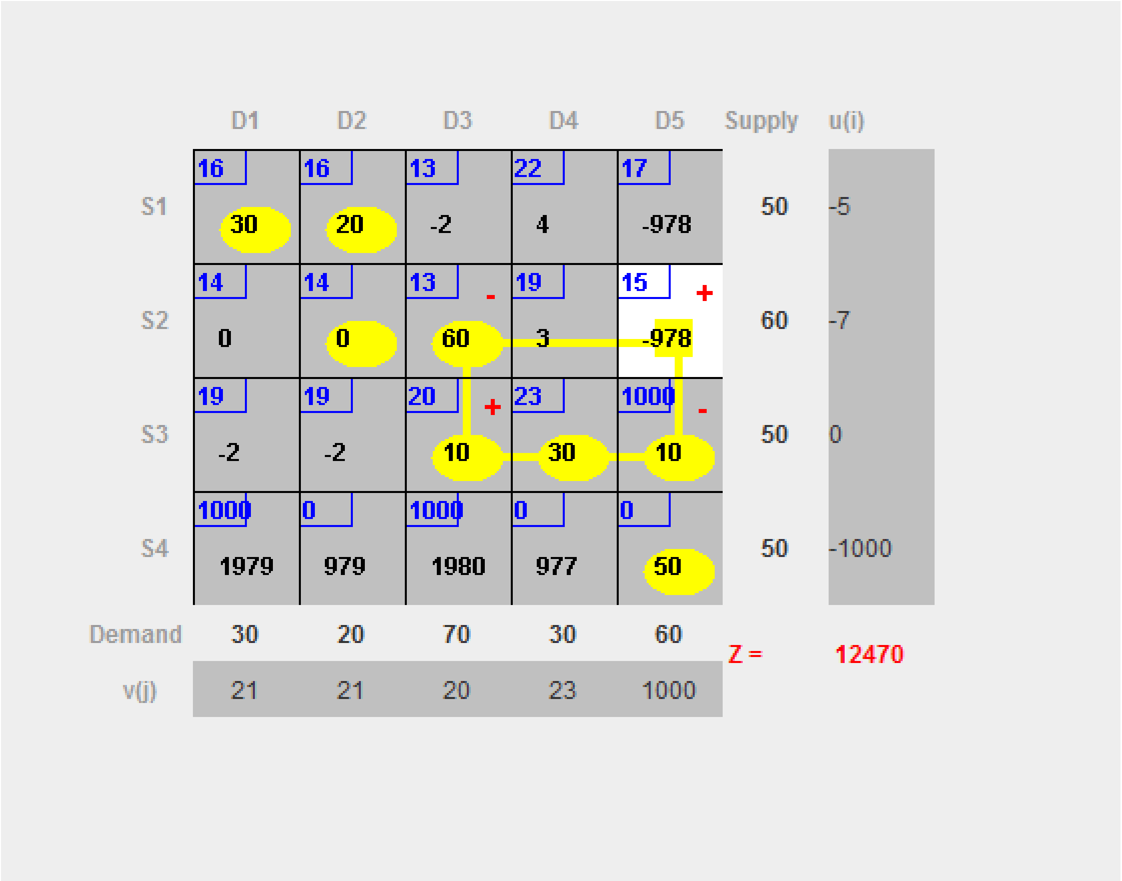
- Set u_3=0, then the rest of the u and v variables are uniquely determined.
- After this we can fill in the “top row” entries of the nonbasic variables (in the grey squares).
- There are two “most negative” entries: the two -978 squares. We can choose either as the entering basic variable.

“Highfive” vintage smiley

A NON-DEGENERATE PIVOT IN THE TRANSPORTATION SIMPLEX METHOD. OBJECTIVE FUNCTION IMPROVES!

“Facepalm” vintage smiley

A DEGENERATE PIVOT, CAUSED BY A BASIC VARIABLE CARRYING ZERO VALUE THAT NEEDS TO DECREASE WHEN THE ENTERING (CURRENTLY NON-BASIC) VARIABLE WANTS TO INCREASE ABOVE ZERO. OBJECTIVE FUNCTION IS NOT GOING TO IMPROVE



	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 / 30	16 / 20	13 / -2	22 / 4	17 / -978	50	-5
S2	14 / 0	14 / 0	13 - / 60	19 / 3	15 + / -978	60	-7
S3	19 / -2	19 / -2	20 + / 10	23 / 30	1000 - / 10	50	0
S4	1000 / 1979	0 / 979	1000 / 1980	0 / 977	0 / 50	50	-1000
Demand	30	20	70	30	60	Z =	12470
v(j)	21	21	20	23	1000		

- (S2,D5) basic: it will have a value of 10, which is the Min (basic variables that have to decrease).

- (S3,D5) will leave. The Z value will improve (i.e., get lower).

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 30	16 20	13 -2	22 4	17 0	50	2
S2	14 0	14 0	13 50	19 3	15 10	60	0
S3	19 -2	19 -2	20 20	23 30	1000 978	50	7
S4	1000 1001	0 1	1000 1002	0 -1	0 50	50	-15
Demand	30	20	70	30	60	Z =	2690
v(j)	14	14	13	16	15		

- Determine if we have reached optimality

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 30	16 20	13 -2	22 4	17 0	50	2
S2	14 0	14 - 0	13 + 50	19 3	15 10	60	0
S3	19 -2	19 + -2	20 - 20	23 30	1000 978	50	7
S4	1000 1001	0 1	1000 1002	0 -1	0 50	50	-15
Demand	30	20	70	30	60		
v(j)	14	14	13	16	15		

Z = 2690

- (S3,D2) becomes basic.
  This is going to be a degenerate pivot, can you see why?
  The Z value is not going to change…

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 30	16 20	13 -4	22 2	17 -2	50	-3
S2	14 2	14 2	13 50	19 3	15 10	60	-7
S3	19 0	19 0	20 20	23 30	1000 978	50	0
S4	1000 1003	0 3	1000 1002	0 -1	0 50	50	-22
Demand	30	20	70	30	60	Z =	2690
v(j)	19	19	20	23	22		

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 / 30	16 / 20	13 / -4	22 / 2	17 / -2	50	-3
S2	14 / 2	14 / 2	13 / 50	19 / 3	15 / 10	60	-7
S3	19 / 0	19 / 0	20 / 20	23 / 30	1000 / 978	50	0
S4	1000 / 1003	0 / 3	1000 / 1002	0 / -1	0 / 50	50	-22
Demand	30	20	70	30	60		
v(j)	19	19	20	23	22		

Z = 2690

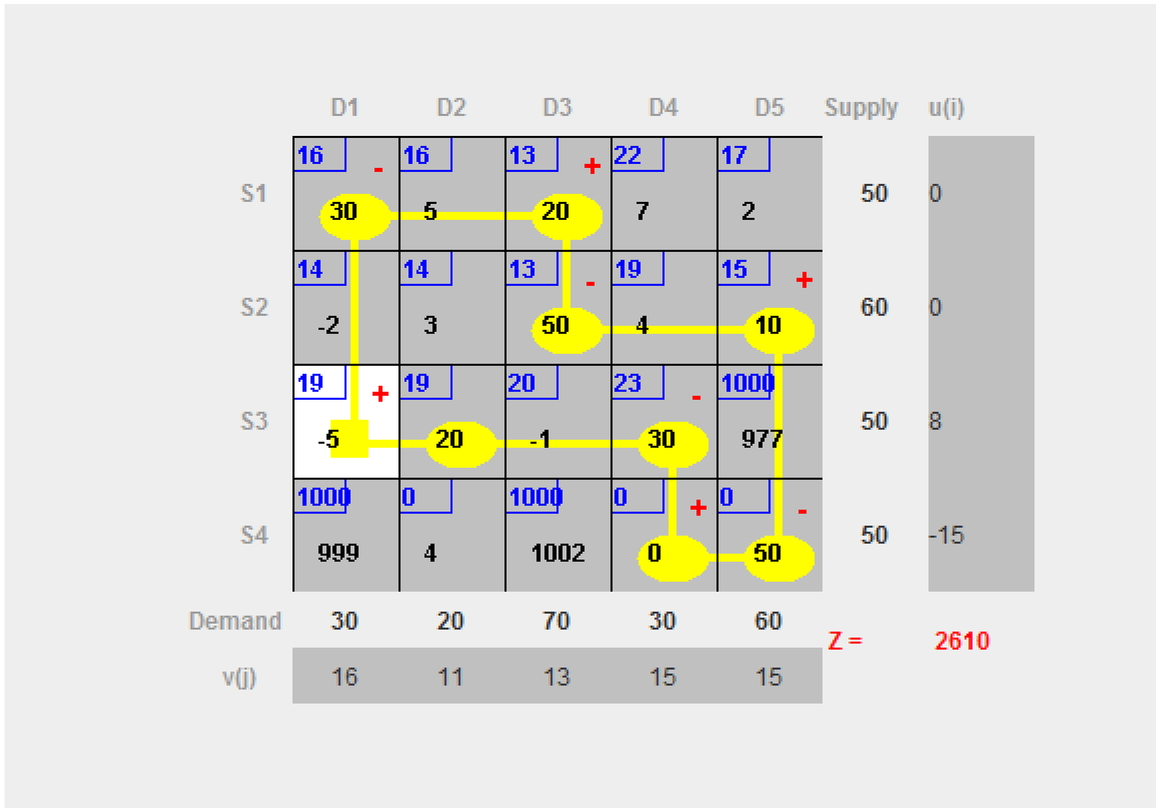
	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 30	16 0	13 20	22 2	17 2	50	0
S2	14 -2	14 -2	13 50	19 -1	15 10	60	0
S3	19 0	19 20	20 4	23 30	1000 982	50	3
S4	1000 999	0 -1	1000 1002	0 -5	0 50	50	-15
Demand	30	20	70	30	60	Z =	2610
v(j)	16	16	13	20	15		

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 / 30	16 / 0 -	13 / 20 +	22 / 2	17 / 2	50	0
S2	14 / -2	14 / -2	13 / 50 -	19 / -1	15 / 10 +	60	0
S3	19 / 0	19 / 20 +	20 / 4	23 / 30 -	1000 / 982	50	3
S4	1000 / 999	0 / -1	1000 / 1002	0 / -5 +	0 / 50 -	50	-15
Demand	30	20	70	30	60	Z =	2610
v(j)	16	16	13	20	15		

- (S4,D4) basic. This will again be degenerate. The Z value is not going to change.

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 30	16 5	13 20	22 7	17 2	50	0
S2	14 -2	14 3	13 50	19 4	15 10	60	0
S3	19 -5	19 20	20 -1	23 30	1000 977	50	8
S4	1000 999	0 4	1000 1002	0 0	0 50	50	-15
Demand	30	20	70	30	60		
v(j)	16	11	13	15	15		

Z = 2610



- (S3,D1) basic. We can set its value to 30.
  The Z Value will change by (30*-5)= -150, to 2460.

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 / 0	16 / 0	13 / 50	22 / 7	17 / 2	50	0
S2	14 / -2	14 / -2	13 / 20	19 / 4	15 / 40	60	0
S3	19 / 30	19 / 20	20 / 4	23 / 5	1000 / 982	50	3
S4	1000 / 999	0 / -1	1000 / 1002	0 / 30	0 / 20	50	-15
Demand	30	20	70	30	60		
v(j)	16	16	13	15	15		

Z = 2460

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 / 0 -	16 / 0	13 / 50 +	22 / 7	17 / 2	50	0
S2	14 / -2	14 / -2 +	13 / 20 -	19 / 4	15 / 40	60	0
S3	19 / 30 +	19 / 20 -	20 / 4	23 / 5	1000 / 982	50	3
S4	1000 / 999	0 / -1	1000 / 1002	0 / 30	0 / 20	50	-15
Demand	30	20	70	30	60		
v(j)	16	16	13	15	15		

Z = 2460

	D1	D2	D3	D4	D5	Supply	u(i)
S1	16 / 2	16 / 2	13 / 50	22 / 7	17 / 2	50	0
S2	14 / 0	14 / 0	13 / 20	19 / 4	15 / 40	60	0
S3	19 / 30	19 / 20	20 / 2	23 / 3	1000 / 980	50	5
S4	1000 / 1001	0 / 1	1000 / 1002	0 / 30	0 / 20	50	-15
Demand	30	20	70	30	60	Z =	2460
v(j)	14	14	13	15	15		

- Finished because no more negative entries in the grey squares!
- Optimum is 2460, values of basic variables are in the yellow circles, all other variables have value 0.

Comments

- The Northwest Corner Rule is simple and easy to apply, but you might start off quite far from optimality. The other starting rules mentioned in the book take longer to apply, but tend to give you a much better starting point.

Comments

- The Northwest Corner Rule is simple and easy to apply, but you might start off quite far from optimality. The other starting rules mentioned in the book take longer to apply, but tend to give you a much better starting point.

- We saw here the phenomenon of **degeneracy**.
  A BFS is *degenerate* if at least one of the basic variables are zero. (Remember, nonbasic variables are *always* zero, while basic variables *might* be zero). As in the general Simplex Method this can be a cause for concern.

Comments

- A common problem with degeneracy is that it can cause iterations to occur where the Z value does not increase.

Comments

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- Relatedly, we saw that degeneracy can *slightly* confuse the optimality test. Namely, we saw a BFS where there was a ? with negative value, but such that the BFS was actually already at optimality. Don't worry, the golden rule still holds: *IF* there are no negative values *THEN* we are definitely at optimality.

Comments

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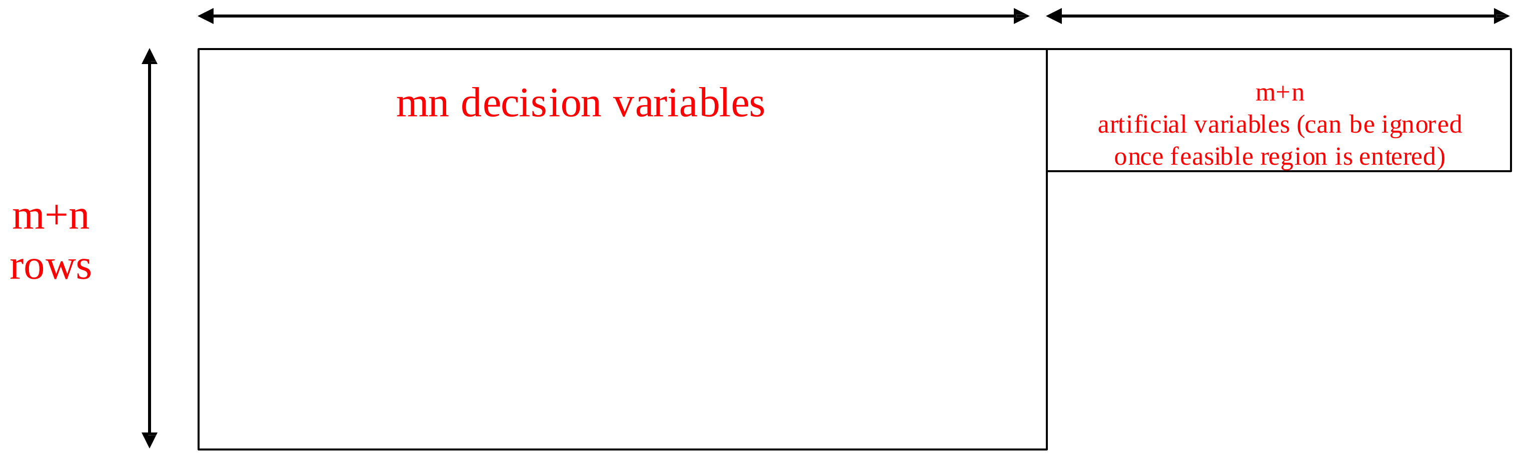
- Relatedly, we saw that degeneracy can *slightly* confuse the optimality test. Namely, we saw a BFS where there was a ? with negative value, but such that the BFS was actually already at optimality. Don't worry, the golden rule still holds: *IF* there are no negative values *THEN* we are definitely at optimality.

- In the worst case degeneracy can cause *cycling*…you return to a previous BFS without having improved the Z value, and thus get stuck in an infinite loop. But in practice this rarely happens and you can, if you want, guard against it.

Pros

Where does the TSM save most time, compared to the normal Simplex method?

- In the pivoting step (i.e. updating the tableaux after having identified the leaving and entering basic variables).
- If you solve a Transportation problem with the normal simplex method, each pivot will require roughly the following number of arithmetic operations: $\boldsymbol{O}\mathbf{(\ (m+n)mn\ ) =} \boldsymbol{O}\mathbf{(max(m^2n,\ mn^2)).}$



Pros

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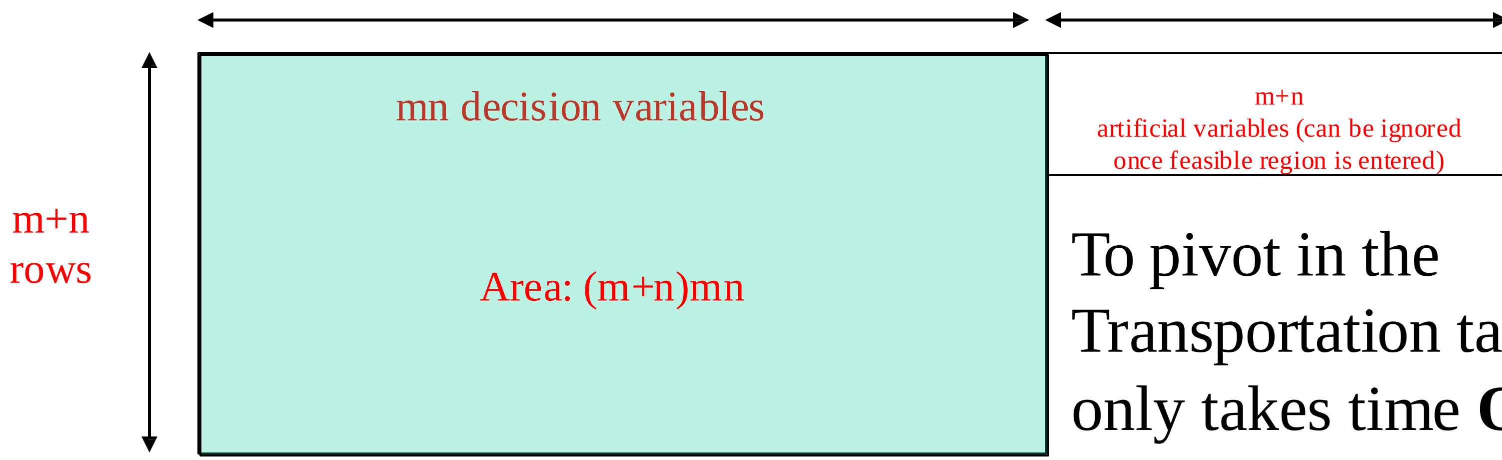

To pivot in the Transportation tableaux only takes time **O(mn).**